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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S2 (PT) (R,S) Examination June 2023 (2019 Scheme)



Course Code: EET204

Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|---|---|
| 1 | Calculate the gradient of a scalar field, $V=4xz^2+3yz$ at (1,1,1) | 3 |
| 2 | Check whether the vector field $A=2z\sin\phi\mathbf{a}_\rho + \rho z\cos\phi\mathbf{a}_\phi + \rho^2\sin\phi\mathbf{a}_z$ is irrotational or not. | 3 |
| 3 | Using Gauss's law, derive the expression for electric field intensity due to a point charge | 3 |
| 4 | What is an electric dipole? Define dipole moment. | 3 |
| 5 | Calculate the magnetic field intensity at (0,1,0) if the entire z axis carries a current of 2A in the positive z direction. | 3 |
| 6 | Distinguish between displacement current density and conduction current density | 3 |
| 7 | A uniform plane wave in free space is given by $\mathbf{H}=0.1\cos(2\times 10^8t-\beta x)\mathbf{a}_y$ A/m. Calculate the value of β . | 3 |
| 8 | Calculate the intrinsic impedance of a lossless dielectric if its relative permeability is 8 and relative permittivity is 2. | 3 |
| 9 | Define standing wave ratio of a transmission line | 3 |
| 10 | State the condition for a transmission line to be lossless. How does the propagation constant of a lossless line is related to the line parameters? | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- | | | |
|----|--|----|
| 11 | a) Transform the vector $\mathbf{E}=z\mathbf{a}_y$ to spherical coordinates | 4 |
| | b) Verify divergence theorem for the vector field, $\mathbf{A}=\rho^2z\mathbf{a}_\rho + 3\rho z\sin\phi\mathbf{a}_\phi + \rho\mathbf{a}_z$ in the region bounded by a cylinder defined by $0\leq\rho\leq 3, 0\leq\phi\leq 2\pi, 0\leq z\leq 4$. | 10 |
| 12 | a) A point P in space is given by (2, 4, 7) in cartesian system of coordinates. What are the coordinates of P in cylindrical system? | 4 |

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- b) Verify Stoke's theorem for the vector field, $\mathbf{F} = x^2y\mathbf{a}_x - y\mathbf{a}_y$ over the contour 10
formed by rectangle with corners (0,0,0), (2,0,0), (2,1,0) and (0,1,0).

Module -2

- 13 a) A point charge of 50pC is located at (2,1,3) while the x-axis carries a charge of 9
2nC/m. The plane $z=3$ carries 10nC/m² charge. Calculate the electric field
intensity at (2,2,2). 5
- b) Derive the expression for capacitance of a coaxial cable 5
- 14 a) A point charge of 4nC is located at (-4,5,0) while the line $x=1, z=1$ carries 9
uniform charge 3nC/m. If $V=0V$ at origin, find V at point A (3,2,1). 9
- b) Using Gauss's law, derive an expression for electric flux density due to a 5
uniformly charged sphere

Module -3

- 15 a) Derive the expression for magnetic field intensity on the axis of a circular loop 10
carrying current I .
- b) Given the magnetic vector potential $\mathbf{A} = \frac{-\rho^2}{4} \mathbf{a}_z$ Wb/m, calculate the magnetic 4
flux density at the point (3,4,0).
- 16 a) Derive the magneto-static boundary conditions at the interface between two 10
different magnetic media.
- b) Explain the modification required to make Ampere's circuital law consistent 4
under a time varying field.

Module -4

- 17 a) Derive electromagnetic wave equations from Maxwell's equations 10
- b) At 50MHz, a lossy dielectric material is characterized by permittivity, $\epsilon = 3.6\epsilon_0$, 4
and permeability $\mu = 2.1\mu_0$ and conductivity, $\sigma = 0.08S/m$. Compute propagation
constant of the medium.
- 18 a) Verify Poynting's theorem for power flow in a coaxial cable 9
- b) Given that $\mathbf{H} = 0.5e^{-0.1x} \sin(2\pi \times 10^6 t - 2x) \mathbf{a}_z$ A/m, calculate the following 5
(i) Attenuation constant (ii) Phase constant (iii) Angular frequency (iv) Direction
of wave propagation (v) Time period

Module -5

- 19 a) Derive wave equations for a transmission line. 10
- b) A lossless transmission line having characteristic impedance of 120Ω is 4

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operating at angular frequency of 5×10^8 rad/s. If the wave velocity on the line is 2.4×10^8 m/s, Calculate inductance and capacitance per metre length of the line

- 20 a) Define the following terms with reference to wave propagation over a transmission line
- (i) Propagation constant (ii) intrinsic impedance (iii) voltage reflection coefficient
- b) The characteristic impedance of a lossless transmission line operating at 80MHz is 72Ω . If the inductance per metre length, $L=0.5 \mu\text{H/m}$, find (i) capacitance per metre length (ii) velocity of wave propagation (iii) phase constant (iv) voltage reflection coefficient if the line is terminated with a load of 60Ω
